

Basic Counting

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Choosing two representatives

A club has 40 members and must choose two representatives.
Choosing a first member and then a second gives

$$40 \cdot 39 = 1560$$

ordered choices.

But the pair consisting of Alice and Bob is the same pair whether Alice or Bob is selected first.

Each unordered pair has been counted twice, so the number of pairs is

$$\frac{40 \cdot 39}{2} = 780.$$

Product rule

Suppose a construction consists of two successive choices.

If the first choice has m possibilities and, for every first choice, the second has n possibilities, then the complete construction has

$$mn$$

possibilities.

This is the **product rule**.

More generally, successive choice counts $n_1, n_2, \ldots, n_k$ give

$$n_1 n_2 \cdots n_k$$

possible sequences of choices.

Ordered and unordered choices

The product rule counts the procedure that was described.

If different procedures produce the same object, division may be needed to correct overcounting.

For two distinct representatives, every unordered pair has exactly two orderings:

$$(a, b) \quad \text{and} \quad (b, a).$$

Counting requires identifying both what is being constructed and when two constructions represent the same outcome.

Sum rule

Suppose a set of outcomes is divided into two disjoint classes.
If the first class contains m outcomes and the second contains n , then there are

$$m + n$$

outcomes in total.

This is the **sum rule**.

The classes must be disjoint. If they overlap, simply adding their sizes counts the overlap more than once.

Cardinality

A **set** is a collection of distinct objects called its elements.

The **cardinality** of a finite set A , written $|A|$, is the number of elements in A .

For example,

$$|\{2, 3, 5, 7, 11\}| = 5.$$

The **empty set**, written $\emptyset$, has no elements, so $|\emptyset| = 0$.

Subsets

A set A is a **subset** of B , written $A \subseteq B$, if every element of A is also an element of B .

For example,

$$\{1, 3\} \subseteq \{1, 2, 3, 4\}.$$

Every set is a subset of itself, and the empty set is a subset of every set:

$$A \subseteq A, \quad \emptyset \subseteq A.$$

A subset is determined only by which elements it contains, not by their order.

Set operations

For sets A and B :

- ▶ $A \cup B$ is the **union**: elements in A or B , or both;
- ▶ $A \cap B$ is the **intersection**: elements in both A and B ;
- ▶ $A \setminus B$ is the **difference**: elements in A but not in B .

These operations describe how counted classes overlap.

Two-set inclusion–exclusion

Adding $|A|$ and $|B|$ counts every element of $A \cap B$ twice.
Subtracting the intersection once corrects the overcount:

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

This is the two-set **inclusion–exclusion principle**.
When $A \cap B = \emptyset$, it reduces to the sum rule.

Strings using three symbols

Let a **string of length five** be an ordered sequence of five symbols chosen from $\{A, B, C\}$.

There are

$$3^5 = 243$$

strings in total by the product rule.

How many strings use all three symbols at least once?

It is easier to count strings that omit at least one symbol and subtract them from the total.

The sets of missing symbols

Let U be the set of all length-five strings over $\{A, B, C\}$.
Define

$$M_A = \{\text{strings in } U \text{ that contain no } A\},$$

$$M_B = \{\text{strings in } U \text{ that contain no } B\},$$

$$M_C = \{\text{strings in } U \text{ that contain no } C\}.$$

Each set has size 2^5 , because only two symbols remain available.
Each pairwise intersection has size 1: a string missing two symbols must use the remaining symbol in all five positions.

Three-set inclusion–exclusion

For finite sets A, B, C ,

$$\begin{aligned}|A \cup B \cup C| &= |A| + |B| + |C| \\ &\quad - |A \cap B| - |A \cap C| - |B \cap C| \\ &\quad + |A \cap B \cap C|.\end{aligned}$$

Single sets are added, pairwise overlaps are subtracted, and the triple overlap is added back.

The alternating signs correct repeated counting.

Strings containing every symbol

Apply inclusion–exclusion to the missing-symbol sets:

$$\begin{aligned}|M_A \cup M_B \cup M_C| &= 3 \cdot 2^5 - 3 \cdot 1^5 + 0 \\ &= 93.\end{aligned}$$

The triple intersection is empty because a length-five string cannot omit all three available symbols.

Therefore the number using every symbol is

$$3^5 - 93 = 150.$$

Small examples

Let A be a finite set. A subset is formed by deciding which elements of A to include.

The first examples are:

$ A $	0	1	2	3
number of subsets	1	2	4	8

This suggests that an n -element set has 2^n subsets.

The product rule proves the formula once we specify the choices precisely.

Choosing membership

Let

$$A = \{a_1, a_2, \dots, a_n\}.$$

To construct a subset $S \subseteq A$, consider each element a_i and choose:

include a_i or exclude a_i .

There are two choices at each of n steps, so the product rule gives

$$2 \cdot 2 \cdots 2 = 2^n$$

choice sequences.

Number of subsets

Theorem

A set with n elements has exactly

$$2^n$$

subsets.

This includes the empty set and the entire set.

For $n = 0$, the formula gives $2^0 = 1$: the empty set has the single subset $\emptyset$.

Summary

- ▶ The product rule counts successive choices; the sum rule counts disjoint cases.
- ▶ Overcounting occurs when several constructions represent the same outcome.
- ▶ Inclusion–exclusion corrects counts of overlapping sets.
- ▶ A subset is determined by one include/exclude choice per element.
- ▶ An n -element set has 2^n subsets.